

**Prompt4-1: using the finite difference method**

Solve this Timoshenko beam forward problem from scratch using the finite difference method.

Problem details:

A single Timoshenko beam with a dimensionless spatial domain  $x \in [0, \pi]$  (length direction) and time domain  $t \in [0, 1]$ . The beam follows linear dynamics with transverse shear deformation considered, governed by the dimensionless coupled partial differential equations:

$$\begin{cases} \theta_{tt} - \theta_{xx} + (\theta - \omega_x) = 0 \\ \omega_{tt} + (\theta - \omega_x)_x = g(x, t) \end{cases}$$

Where:

- $\omega(x, t)$  = transverse displacement,  $\theta(x, t)$  = cross-sectional rotation,
- $g(x, t) = \cos(t) - \frac{\pi}{2} \sin(x) \cos(t)$ , applied external force
- All material/geometric parameters are dimensionless and set to 1.

Initial conditions:

- At  $t = 0$ :  $\omega(x, 0) = \frac{\pi}{2} \sin(x)$ ,  $\omega_t(x, 0) = 0$
- At  $t = 0$ :  $\theta(x, 0) = \frac{\pi}{2} \cos(x) + (x - \frac{\pi}{2})$ ,  $\theta_t(x, 0) = 0$

Boundary conditions:

- At  $x = 0$  and  $x = \pi$ :  $\omega(0, t) = \omega(\pi, t) = 0$
- At  $x = 0$  and  $x = \pi$ :  $\theta(0, t) = \theta(\pi, t) = 0$

Calculations required:

- Output the spatial distribution of Transverse displacement  $\omega(x, t)$  and Cross-sectional rotation  $\theta(x, t)$ , for  $t \in [0, 1]$ .

Programming language: Python 3.10.

Permitted libraries: NumPy 2.1.1, SciPy 1.14.1, Matplotlib 3.9.2.